Experiment 3: SMO Trafficking Time Course

Purpose

Measure how quickly SMO leaves the primary cilium after AP21967-induced recruitment of MGRN1, and whether the amount of SMO in the cell as a whole also changes. Ciliary and whole-cell SMO are measured from the same images across a six-hour time course. The time-resolved measurements also provide the data the kinetic model needs.

Experimental design

- Ciliated NIH/3T3 cells carrying the direct MGRN1 recruitment system are treated with AP21967 or ethanol, then fixed and imaged at baseline and at 30, 60, 120, 240 and 360 minutes. Separate wells are fixed at each timepoint, which avoids continuous live-cell imaging.
- SMO is read through a fluorescent SMO readout together with a ciliary marker. For each cell, background-corrected SMO fluorescence is measured in the cilium and in a whole-cell region, using the same imaging channel and acquisition settings. Cilium length and area are tracked so that a change in cilium size is not mistaken for a change in SMO.
- The first pass uses a single AP21967 concentration against the ethanol. If this produces a measurable response, key timepoints are repeated at further concentrations to test whether depletion depends on dose. A longer series of six to eight concentrations follows if the model needs it, in particular to test whether the response plateaus or falls away at high concentrations.
- Cell health and the proportion of cells with a primary cilium are monitored in every condition.

Constructs and conditions

Condition	SMO construct	Recruited protein	AP21967
Direct MGRN1 recruitment	SMO_DmrC_T2098L_CeGFP	MGRN1_DmrA_His6	ethanol and AP21967
Catalytically inactive MGRN1	SMO_DmrC_T2098L_CeGFP	MGRN1_C279A_C282A_DmrA_His6	ethanol and AP21967
Recruitment-deficient	SMO_DmrC_T2098L_CeGFP	DmrA_mCherry_proximity_only_control	ethanol and AP21967
Dimerizer only	SMO_DmrC_T2098L_CeGFP	None	ethanol and AP21967

Constructs are the Experiment 1 constructs, prepared as described there.

Materials and equipment

- NIH/3T3 cells and complete medium (DMEM with high glucose, 10% FBS, GlutaMAX, 1 mM sodium pyruvate, 1× MEM non-essential amino acids, 1× penicillin-streptomycin)
- Constructs above, purified by midiprep (Experiment 1)
- AP21967 (Takara A/C Heterodimerizer) and ethanol
- 24-well plates with glass coverslips
- Transfection reagent: X-tremeGENE 9 in Opti-MEM
- Low-serum medium (complete medium with 0.5% FBS)
- Chilled 4% paraformaldehyde in PBS; blocking buffer of 1% horse serum and 0.1% Triton X-100 in PBS
- Primary antibodies: ARL13B (1:1000); chicken anti-GFP (1:1000) to enhance the SMO-GFP signal. Alexa Fluor secondary antibodies.
- Mounting medium with DAPI
- Fluorescence microscope with a 60× oil objective, and Fiji for analysis

Procedure: cell preparation and ciliation

1. Plate 100,000 NIH/3T3 cells per well on glass coverslips in a 24-well plate, in 500 µL of antibiotic-free complete medium.
2. Express the recruitment system by transient transfection of the constructs, as in Experiment 1. Transfect about 1 hour after plating.
3. Once the cells are confluent, switch to low-serum medium (complete medium with 0.5% FBS) for 24 hours to induce primary cilia.

Procedure: AP21967 time course

1. Dilute AP21967 in medium to a single working concentration [500 nM, as in Experiment 1], and prepare ethanol at the same final concentration.
2. Use separate wells for each timepoint: baseline (0 minutes), 30, 60, 120, 240 and 360 minutes, for AP21967 and for ethanol.
3. Add AP21967 after ciliation, in reverse order, starting with the 360-minute wells, so that every well is fixed at the same time.
4. Fix all wells together: chilled 4% paraformaldehyde in PBS for 10 minutes on ice, then three 5-minute rinses in PBS.

Procedure: staining and imaging

1. Block the cells for 1 hour in 1% horse serum and 0.1% Triton X-100 in PBS.
2. Incubate with primary antibodies overnight at 4 °C, then with secondary antibodies for 1 hour at room temperature, all diluted in blocking buffer. Use a far-red secondary antibody for the ciliary marker, so it does not overlap the SMO GFP channel or the mCherry in the recruitment-deficient control.
3. Mount the coverslips with an antifade mounting medium containing DAPI.
4. Image with a 60× oil immersion objective (NA 1.42), acquiring z-stacks about 4 µm thick, with the same acquisition settings for every well within the experiment.

Procedure: quantification

1. Open the stacks in Fiji as maximum intensity projections.
2. Build ciliary masks from the ARL13B signal, only in transfected (GFP-positive) cells, and apply them to the SMO channel to measure ciliary SMO fluorescence.
3. Draw a whole-cell mask for each transfected cell and measure SMO fluorescence in the same channel and from the same images; where possible, also measure the non-ciliary part of the cell.

4. For each image, subtract the mean intensity of a cell-free background region.
5. Record cilium length and area, and the proportion of ciliated cells.
6. Analyse about 20-50 cilia per condition.
7. Normalise each measurement to its baseline and report ciliary SMO, whole-cell SMO and their ratio at each timepoint.

Procedure: dose series (if the single-dose response is measurable)

1. Repeat key timepoints across a series of AP21967 concentrations and ethanol ethanol. The manufacturer recommends testing within 0.05–500 nM; one possible series is 0.05, 0.5, 1.6, 5, 16, 50, 160 and 500 nM.

Controls

- ethanol ethanol gives the baseline trafficking profile.
- Catalytically inactive MGRN1 tests whether the response requires ligase activity.
- A recruitment-deficient condition tests whether the response requires the engineered interaction at all.
- A dimerizer-only control catches anything AP21967 does by itself, including any effect on mTOR signalling or on the stability of the DmrC fusion.
- Cell health and the proportion of ciliated cells are monitored in every condition.

Expected readout and analysis

- If recruitment drives SMO down, ciliary SMO fluorescence should fall after AP21967 relative to ethanol. The whole-cell measurement at the same time shows what that fall means.
- Ciliary signal down with little change in the whole cell would be consistent with SMO being redistributed or leaving the cilium. Both falling together would be consistent with less SMO in the cell overall, although confirming degradation would require a biochemical abundance assay rather than imaging. No change, with recruitment verified, would indicate that recruitment does not produce a measurable trafficking response under these conditions.
- These measurements let the modelling team estimate how fast and how far ciliary SMO falls and constrain a model of how SMO is distributed between compartments. They cannot separate ciliary entry and exit rate constants on their own; that would need another method, such as fluorescence recovery or photoconversion.

References

- Parashara et al., 2025. The E3 ubiquitin ligase MGRN1 targets melanocortin receptors MC1R and MC4R via interactions with transmembrane adapters. J Cell Sci 138, jcs264084. https://doi.org/10.1242/jcs.264084 (cell culture, ciliation, immunofluorescence and ciliary quantification methods)
- Takara Bio, A/C Heterodimerizer (identical to AP21967), recommended concentration range: https://takarabio.com/x9691.xml